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Backpack And Backpack Connecting Components

Abstract

This utility model discloses a backpack and a set of backpack connecting components. The backpack comprises a backpack body, a connecting component A set on the left side of the backpack body, a connecting component B set on the right side of the backpack body, and a connecting component C set on the bottom of the backpack body. Connecting component A is used for detachable connection with the component a set on the left side of the bicycle seat, connecting component B is used for detachable connection with the component b set on the right side of the bicycle seat, and connecting component C is used for detachable connection with the component c set on the bicycle seat post. This backpack can be used as a regular backpack and can also be used with a bicycle equipped with the connecting components for quick disassembly and assembly.

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Background/Summary

TECHNICAL FIELD

[0001] This utility model pertains to the field of backpack technology, specifically involving a backpack and a set of backpack connecting components.

BACKGROUND

[0002] For carrying loads on bicycles (including manual and electric bicycles) located behind the seat post, existing solutions typically involve installing a rear rack for holding items behind the seat post, which can be paired with bags to facilitate item loading. However, due to issues such as increased weight, impact on aesthetics, and added noise caused by installing a rear rack, it affects the riding experience, leading some cyclists to be reluctant to add a rear rack.

[0003] Currently, there is also a type of tail-specific bag that is directly fixed in the space behind the seat post using methods like strapping or brackets. Since this bag is designed for specific use, users don't use it for everyday purposes when not cycling. Therefore, when stopping cycling, items from the specialized bag need to be transferred to a regular backpack. Additionally, these tail-specific bags often adopt a unique shape with a wider top and a narrower bottom to avoid interference with the rider's inner thighs/rear side. However, this design results in low space utilization when loading items.

SUMMARY

[0004] In response to the shortcomings of existing technologies, this utility model provides a backpack and a set of backpack connecting components. It can be used as a regular backpack in daily life and can be combined with a bicycle equipped with the connecting component for quick disassembly and assembly. During cycling, it avoids interference with the rider's inner thighs/rear side and enhances the cycling experience. After cycling, it allows for quick disassembly and carrying without the need to transfer items, optimizing the overall cycling experience.

[0005] The technical solution of this utility model is as follows:

[0006] On one hand, it provides a backpack comprising a backpack body, connecting component A set on the left side of the backpack body, connecting component B set on the right side of the backpack body, and connecting component C set on the bottom of the backpack body. Connecting component A is used for detachable connection with the component a set on the left side of the bicycle seat, connecting component B is used for detachable connection with the component b set on the right side of the bicycle seat, and connecting component C is used for detachable connection with the component c set on the bicycle seat post. The connections between connecting component A and component a, connecting component B and component b, and connecting component C and component c collectively constitute the support structure of the backpack body, and the connection between connecting component C and component c creates spacing between the backpack body and the seat post.

[0007] Preferably, the bottom of the backpack body is fixed with a base, and the base has a mounting slot with one end open to the outside, facing the seat post. The mounting slot is used to accommodate connecting component C.

[0008] Preferably, connecting component C is specifically a sleeve, and connecting component c includes a support rod and a clamping hoop. The clamping hoop tightens around the seat post, one end of the support rod is connected to the clamping hoop, the support rod matches the sleeve, and the support rod can be inserted into the sleeve and locked by a pin.

[0009] Preferably, pivot shaft X is set inside the mounting slot near the seat post, and one end of the sleeve can be rotatably sleeved on pivot shaft X. A limiting cam protrusion is set at the far end away from the seat post inside the mounting slot. When the angle between the sleeve and the mounting slot is 0°, the limiting cam protrusion is used to limit the sleeve. When the angle is 180°, at least $\frac{1}{3}$ of the length of the sleeve remains inside the mounting slot.

[0010] Preferably, the sleeve is fixed in the mounting slot, and one end of the support rod is hinged to the clamping hoop. The support rod can rotate relative to the seat post in the range of 0-90°, and

the support rod can be locked at 0° and 90° relative to the seat post.

[0011] Preferably, the connections between connecting component A and component a, and connecting component B and component b, can use snap-on, hook-on, or strap connections.

[0012] Preferably, the connections between connecting component A and component a, and connecting component B and component b, use snap-on connections. Specifically, connecting component A, connecting component B, component a, and component b all include adjustable straps and buckles. The length of the adjustable straps can be adjusted, and one end of the adjustable strap of connecting component A and connecting component B is connected with the buckles, while the other end is connected to the backpack body. One end of the adjustable strap of component a and component b is connected with the buckles, and the other end is connected to the bicycle seat. The buckles of connecting component A and connecting component b are male, and the buckles of component a and component B are female. The buckles of component a and component b can be docked and fastened.

[0013] On the other hand, this utility model provides a set of backpack connecting components, including any one of the backpacks described above in claims **1-9**, and further comprising connecting component a for connection with the left side of the bicycle seat, connecting component b for connection with the right side of the bicycle seat, and connecting component c for connection with the bicycle seat post.

[0014] The beneficial effects of this utility model are as follows:

[0015] The backpack provided by this utility model achieves quick disassembly and assembly with a bicycle through the detachable connections between connecting component A and component a, connecting component B and component b, and connecting component C and component c. The backpack can be securely fixed on the bicycle through these three points during cycling, eliminating the need for the rider to carry the backpack on his/her back, making cycling more comfortable. After reaching the destination, the backpack can be quickly separated from the bicycle, and the rider does not need to transfer items from any specialized bicycle bag, allowing for quick disassembly and carrying, facilitating the rider to carry the backpack for other activities when not cycling.

[0016] After assembly with a bicycle, the bottom of the backpack and the seat post maintain sufficient distance, preventing interference with the rider's movements and ensuring a comfortable and safe cycling experience.

[0017] The backpack provided by this utility model, through the rotatable and foldable design of the sleeve, allows the sleeve to be folded and stored in the mounting slot on the base when the backpack separated from the bicycle, without affecting the daily use of the backpack. Alternatively, through the rotatable and foldable design of the support rod, the presence of the support rod can be further reduced when the backpack is separated from the bicycle.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0018] For a clearer understanding of specific embodiments of this utility model or the technical solutions in the prior art, brief introductions to the accompanying drawings used in specific embodiments or descriptions of the prior art that are needed are provided below. In all the drawings, similar elements or parts are generally labeled with similar reference symbols. In the drawings, the proportions of various elements or parts may not necessarily be drawn to scale.

[0019] FIG. **1** is a schematic diagram of the structure of the backpack and backpack connecting components provided by Example 1 of this utility model.

[0020] FIG. **2** is a schematic diagram of the structure of the base of the backpack provided by Example 1 of this utility model.

[0021] FIG. 3 is a schematic diagram of the backpack and the bicycle in the connected state provided by Example 2 of this utility model.

[0022] FIG. 4 is a schematic diagram of the backpack and bicycle in the separated state provided by Example 2 of this utility model.

[0023] In the drawings, **1**—backpack body, **11**—connecting component A, **12**—connecting component B, **13**—connecting component C, **131**—sleeve, **14**—base, **141**—mounting slot, **21**—connecting component a, **22**—connecting component b, **23**—connecting component c, **231**—clamping hoop, **232**—support rod, **31**—bicycle seat, **32**—seat post.

DESCRIPTION OF EMBODIMENTS

[0024] In order to facilitate a better understanding of the proposed solution by those skilled in the art, the following detailed and comprehensive description of the technical solutions in the embodiments of this application will be provided in conjunction with the accompanying drawings. It is apparent that the described embodiments are only part of the embodiments of this application, not all embodiments. Based on the embodiments described in this application, all other embodiments obtained by those skilled in the art without engaging in creative work, should fall within the scope of protection of this application.

[0025] It should be noted that the terms “first,” “second,” etc., used in the specification and claims, and the like in the accompanying drawings, are used to distinguish similar objects, and do not necessarily indicate a particular sequence or order. These terms are used interchangeably in appropriate circumstances so that the embodiments described here are understood.

[0026] In this application, the terms “up,” “down,” “inner,” etc., indicating spatial or positional relationships are based on the orientation or position shown in the drawings. These terms are mainly used to better describe the embodiments of this application and are not used to limit the devices, components, or parts indicated to a specific orientation, or to construct and operate in a specific orientation.

[0027] Furthermore, the terms “set,” “provided with,” “connected,” “fixed,” etc., should be broadly interpreted. For example, “connected” can be a fixed connection, a detachable connection, or an integral structure; it can be a mechanical connection or an electrical connection; it can be directly connected, or indirectly connected through an intermediate medium, or internally connected between two devices, components, or parts. For those skilled in the art, these terms can be understood in specific contexts.

[0028] In addition, the meaning of the term “multiple” should be understood as two or more.

[0029] It should be noted that, in the absence of conflict, the embodiments in this application and the features in the embodiments can be combined with each other. The following will detail the embodiments of this application with reference to the drawings.

Example 1

[0030] This embodiment provides a backpack, as shown in FIG. 1, including a backpack body **1**, connecting component **A11** set on the left side of the backpack body **1**, connecting component **B12** set on the right side of the backpack body **1**, and connecting component **C13** set on the bottom of the backpack body **1**. Connecting component **A11** is used for detachable connection with the component **a21** set on the left side of the bicycle seat **31**, connecting component **B12** is used for detachable connection with the component **b22** set on the right side of the bicycle seat **31**, and connecting component **C13** is used for detachable connection with the component **c23** set on the bicycle seat post **32**. The connections between connecting component **A11** and component **a21**, connecting component **B12** and component **b22**, and connecting component **C13** and component **c23** collectively constitute the support structure of the backpack body **1**, and the connection between connecting component **C13** and component **c23** creates spacing between the backpack body **1** and the seat post **32**.

[0031] This design can first firmly fix the backpack on the bicycle during cycling without the need for the rider to carry it on his/her back, making cycling more comfortable. Secondly, there is

enough space between the backpack and the seat post **32** to prevent interference with the rider's movements. Finally, after reaching the cycling destination, the backpack can be quickly separated from the bicycle, allowing for convenient carrying.

[0032] It should be noted that, when the backpack is assembled with the bicycle, the main force is borne by the connecting components on the left and right sides (the connection between connecting component **A11** and connecting component **a21**, and the connection between connecting component **B12** and connecting component **b22**). The purpose of the bottom connecting components (the connection between connecting component **C13** and connecting component **c23**) are to create spacing between the backpack body **1** and the seat post **32** to prevent interference with the rider's inner thighs/rear side and to limit excessive left and right swaying and rotation of the backpack during cycling, improving stability. Additionally, the bottom connecting component can provide auxiliary support during the assembly and disassembly of the backpack.

[0033] In this embodiment, to provide a stable and reliable mounting position for connecting component **C13**, the bottom of the backpack body **1** is fixed with a base **14**, as shown in FIG. 2. The base **14** is preferably made of plastic, and it has a mounting slot **141**. The mounting slot **141** is open to the outside at one end facing the seat post **32** and is used to accommodate connecting component **C13**. The setting of the base **14** also ensures that the backpack body **1** can stand upright on a flat surface even when the backpack separated from the bicycle, making it less prone to tipping. To enhance the aesthetics of the backpack, the outer surface of the base **14** can be covered with fabric.

[0034] In this embodiment, as shown in FIG. 2, connecting component **C13** specifically is a sleeve **131**, and connecting component **c23** specifically includes a support rod **232** and a clamping hoop **231**. The clamping hoop **231** tightens around the seat post **32**, one end of the support rod **232** is connected to the clamping hoop **231**, and the support rod **232** matches the sleeve **131**. The support rod **232** can be inserted into the sleeve **131** and locked by a pin.

[0035] In summary, the connection between connecting component **C13** and connecting component **c23** is achieved by inserting the support rod **232** into the sleeve **131** and locking it with a pin. Connecting component **C13** and connecting component **c23** are preferably made of aluminum alloy. The sleeve **131** has a rounded rectangular cross-section, as shown in FIG. 2, and the cross-sectional shape and size of the support rod **232** are precisely matched to the sleeve **131**. This design provides better strength and rigidity, increases the load-bearing capacity of the sleeve **131**, prevents left and right swaying or rotation of the backpack during cycling, optimizes stability, and the rounded design reduces stress concentration in the sleeve **131** during use, improving durability and service life.

[0036] It should be noted that the connection between connecting component **C13** and connecting component **c23** includes but is not limited to sleeve docking or buckle docking. In some other embodiments, a threaded connection can replace pin locking between sleeve **131** and support rod **232**, where a bearing rotational connection between support rod **232** and clamping hoop **231** can achieve the same effect. In some other embodiments, the positions of sleeve **131** and support rod **232** can be swapped, i.e., connecting component **C13** specifically becomes support rod **232**, while connecting component **c23** specifically includes sleeve **131** and clamping hoop **231**, achieving the same effect.

[0037] Furthermore, near the position of seat post **32** inside mounting slot **141**, pivot shaft X is installed in this embodiment. One end of sleeve **131** is rotatably fitted on pivot shaft X, and a limiting card protrusion is set inside mounting slot **141** at the far end away from seat post **32**. When the angle between sleeve **131** and mounting slot **141** is 0° , the limiting card protrusion is used to limit sleeve **131**. When the angle between sleeve **131** and mounting slot **141** is 180° , at least $\frac{1}{3}$ of the length of sleeve **131** remains inside mounting slot **141**.

[0038] In this embodiment, sleeve **131** adopts a design that can be rotated or say folded. After this design, when the backpack is separated from the bicycle for use, sleeve **131** is completely housed

in mounting slot **141** of base **14** without exposure, and the angle between sleeve **131** and mounting slot **141** is 0° , with the limiting card protrusion limiting sleeve **131**. When it is necessary to use the bicycle to carry the backpack, the user can manually rotate sleeve **131** to overcome the restriction of the limiting card protrusion, continue rotating until the angle between sleeve **131** and mounting slot **141** is 180° . At this point, it can align with support rod **232** for insertion, and subsequently, the connection between them can be locked by a pin. Additionally, as sleeve **131** still has at least $\frac{1}{3}$ of its length inside mounting slot **141** at this time, leverage can be utilized to enhance the stability and load-bearing capacity of the connection. The connections between connecting component **A11** and connecting component **a21**, as well as connecting component **B12** and connecting component **b22**, can serve as auxiliary fixation for the backpack. After reaching the cycling destination, the connections between connecting component **A11** and connecting component **a21**, connecting component **B12** and connecting component **b22**, and connecting component **C13** and connecting component **c23** are released. Then, sleeve **131** is rotated to reset to an angle of 0 with mounting slot **141**, and the backpack can be carried on shoulder.

[0039] It should be noted that, in this embodiment, connecting component **C13** is specifically a straight sleeve **131**. In some other embodiments, connecting component **C13** can also adopt a structure of an isosceles triangle plate+straight sleeve **131**. That is, the straight sleeve **131** is embedded on the perpendicular bisector of the isosceles triangle plate, and the base of the isosceles triangle plate is hinged with mounting slot **141**, achieving the effect of triangular support.

[0040] It should be noted that, in this embodiment, connecting component **C13** achieves the function of being exposed or concealed through rotatable design. In some other embodiments, the connecting component **C13** can also achieves the same function through extension and retraction design. Specifically, the mounting slot may be designed to be enclosed on all sides with only the front end open, allowing the connecting component **C13** to slide within the mounting slot. In this manner, the connecting component **C13** can extend from or retract into the mounting slot.

[0041] It should be noted that, in this embodiment, the specific form of the backpack body includes but is not limited to a backpack with double shoulder straps, a single-shoulder backpack, and a diagonal crossbody backpack.

[0042] In this embodiment, the connection methods between connecting component **A11** and connecting component **a21**, as well as connecting component **B12** and connecting component **b22**, include but are not limited to buckle docking, hook docking, or strapping connection.

[0043] More specifically, as shown in FIG. 1, the connecting component **A11**, connecting component **B12**, connecting component **a21**, and connecting component **b22** of this embodiment all include adjustable straps and buckles. The length of the adjustable strap can be adjusted, and one end of the adjustable strap of connecting component **A11** and the adjustable strap of connecting component **B12** are both connected to the buckle. The other end is connected to the backpack body **1**. The adjustable strap of connecting component **a21** and the adjustable strap of connecting component **b22** are both connected to the buckle, and the other end is connected to the bicycle seat **31**. The buckles of connecting component **A11** and connecting component **b22** are male buckles, and the buckles of connecting component **a21** and connecting component **B12** are female buckles. The buckles of connecting component **a21** and connecting component **b22** can be docked and engaged.

[0044] It should be noted that, after the user docks the buckles of connecting component **A11** with the buckles of connecting component **a21**, and the buckles of connecting component **B12** with the buckles of connecting component **b22**, it is necessary to ensure that the adjustable straps of each connecting component are in a tightened state. This ensures that the main force of the backpack falls on the connecting components on the left and right sides, reducing damage to the bottom connecting components and extending the service life of the connecting components.

[0045] With this design, not only can the connections between connecting component **A11** and connecting component **a21**, as well as connecting component **B12** and connecting component **b22**,

achieve quick installation and removal, but also the connections between connecting component **a21** and connecting component **b22** can be engaged when the backpack is not mounted. This avoids excessive swinging and movement of connecting component **a21** and connecting component **b22** with the bicycle body when the backpack is not in use, preventing interference with cycling.

[0046] This embodiment also provides a set of backpack connecting components, as shown in FIG. 1, including the backpack described above and also including connecting component **a21** for connection with the left side of the bicycle seat **31**, connecting component **b22** for connection with the right side of the bicycle seat **31**, and connecting component **c23** for connection with the bicycle seat post **32**. Connecting component **a21**, connecting component **b22**, and connecting component **c23** are all fixedly connected to the bicycle, and more preferably, they use detachable fixed connections for easy replacement in case of local faults or damage.

Example 2

[0047] This embodiment provides a backpack, as shown in FIGS. 3 to 4, which is generally similar to the backpack provided in Example 1. The difference lies in:

[0048] In this embodiment, sleeve **131** is fixed inside mounting slot **141**. One end of support rod **232** is hinged with clamping hoop **231**, and support rod **232** can rotate in the range of 0-90° relative to seat post **32**. Support rod **232** and seat post **32** can be locked at 0° and 90° angles.

[0049] In this embodiment, sleeve **131** adopts a fixed design, while support rod **232** adopts a design that can be rotated and folded. With this design, when it is necessary to use the bicycle to carry the backpack, the user can rotate support rod **232** to an angle of 90° relative to seat post **32**. At this point, support rod **232** is directly facing sleeve **131**, set horizontally in mounting slot **141** of base **14**. The backpack can be moved to allow sleeve **131** to socket onto support rod **232**, and the connection between them can be locked by a pin. The connections between connecting component **A11** and connecting component **a21**, as well as connecting component **B12** and connecting component **b22**, can serve as fixation and weight-bearing for the backpack. After reaching the cycling destination, the connections between connecting component **A11** and connecting component **a21**, connecting component **B12** and connecting component **b22**, and connecting component **C13** and connecting component **c23** are released. Then, support rod **232** is rotated to reset to an angle of 0 relative to seat post **32**, reducing the presence of support rod **232**, and the backpack can be carried on shoulder.

[0050] In conclusion, the above-described embodiments are only used to illustrate the technical solutions of the present utility model, not to limit them. Although the embodiments have been described in detail with reference to the foregoing embodiments, those skilled in the art should understand that various modifications or substitutions can be made to the technical features described in the foregoing embodiments. These modifications or substitutions do not depart from the spirit and scope of the technical solutions of the present utility model, and should be encompassed by the claims and the description.

Claims

1. A backpack, characterized by comprising: a) a backpack body; b) connecting component **A** positioned on the left side and detachably connected to component **a** on the left side of a bicycle seat; c) connecting component **B** positioned on the right side and detachably connected to component **b** on the right side of the bicycle seat; d) and connecting component **C** positioned at the bottom, and detachably connected to component **c** on the bicycle seat post; and the connections between connecting component **A** and component **a**, connecting component **B** and component **b**, and connecting component **C** and component **c** collectively form the support structure of the backpack body, while the connection between connecting component **C** and component **c** creates spacing between the backpack body and the seat post.

2. According to claim 1, the backpack characterized by: further comprising a base fixed at the

bottom of the backpack body, with a mounting slot that is opened on the base, with one end facing the seat post open, and utilized for accommodating connecting component C.

3. According to claim 2, the backpack characterized by: further comprising a specific sleeve as connecting component C, and comprising a support rod and a clamping hoop as component c with below characteristics: a) the clamping hoop tightens around the seat post; b) one end of the support rod is connected to the clamping hoop; c) the support rod matches the sleeve and can be inserted directly into the sleeve and locked by a pin.

4. According to claim 3, the backpack characterized by: further comprising a pivot shaft, designated as pivot shaft X in this document, near the seat post inside the mounting slot, with one end of the sleeve rotatably fixed on, and a limiting cam protrusion set at the far end away from the seat post inside the mounting slot, making below operation possible: a) when the angle between the sleeve and the mounting slot is 0° , the limiting cam protrusion is used to limit the sleeve; b) and when the angle between the sleeve and the mounting slot is 180° , at least $\frac{1}{3}$ of the length of the sleeve remains inside the mounting slot.

5. According to claim 3, the backpack characterized by: the sleeve is fixed in the mounting slot, and the clamping hoop is hinged to one end of the support rod which can rotate relative to the seat post in the range of $0-90^{\circ}$, and can be locked at 0° and 90° relative to the seat post.

6. According to claim 1, the backpack characterized by: the connections between connecting component A and component a, and connecting component B and component b, can use snap-on, hook-on, or strap connections.

7. According to claim 6, the backpack characterized by: the connections between connecting component A and component a, and connecting component B and component b, use snap-on connections with below characteristics: a) connecting component A, connecting component B, component a, and component b all include adjustable straps and buckles; b) the length of the adjustable straps can be adjusted, one end of the adjustable strap of connecting component A and connecting component B is connected with the buckles, and the other end is connected to the backpack body; c) one end of the adjustable strap of component a and component b is connected with the buckles, and the other end is connected to the bicycle seat; d) the buckles of connecting component A and connecting component b are male, and the buckles of component a and component B are female, making the buckles of component a and component b can be docked and fastened.

8. The connecting component C achieves the function of being exposed or concealed through two manners: extension and retraction design or rotatable design.

9. A backpack connection assembly, characterized in that: it includes the backpack according to claim 1 and further includes a connector a for connecting to the left side of the bicycle seat, and a connector a for connecting to the right side of the bicycle seat. b and the connecting piece c for connecting to the bicycle seat post.
